

Deep Learning Optimisé - Jean Zay

Conclusion



INSTITUT DU
DÉVELOPPEMENT ET DES
RESSOURCES EN
INFORMATIQUE
SCIENTIFIQUE

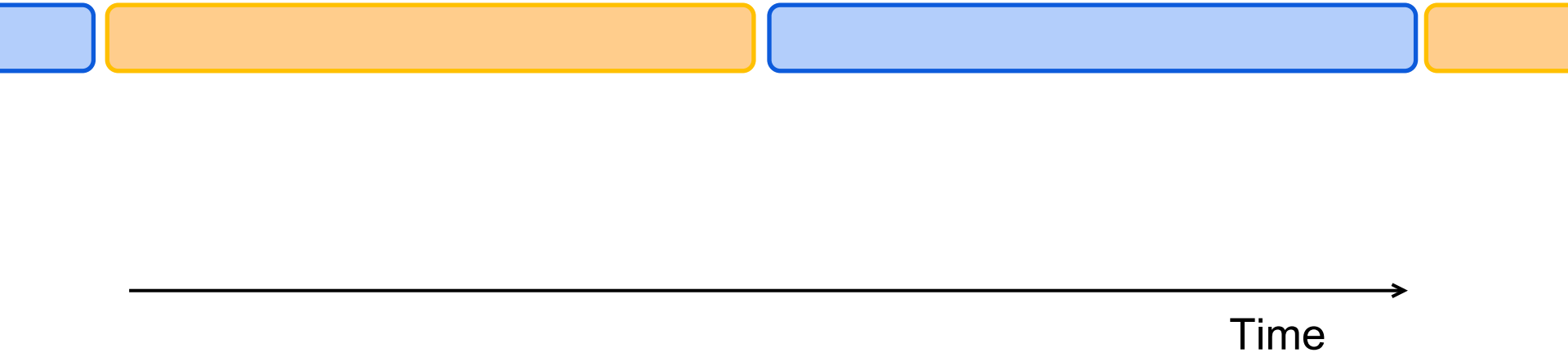
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Conclusion

Synchronous : num_worker = 0

DataLoader

Forward/Backward

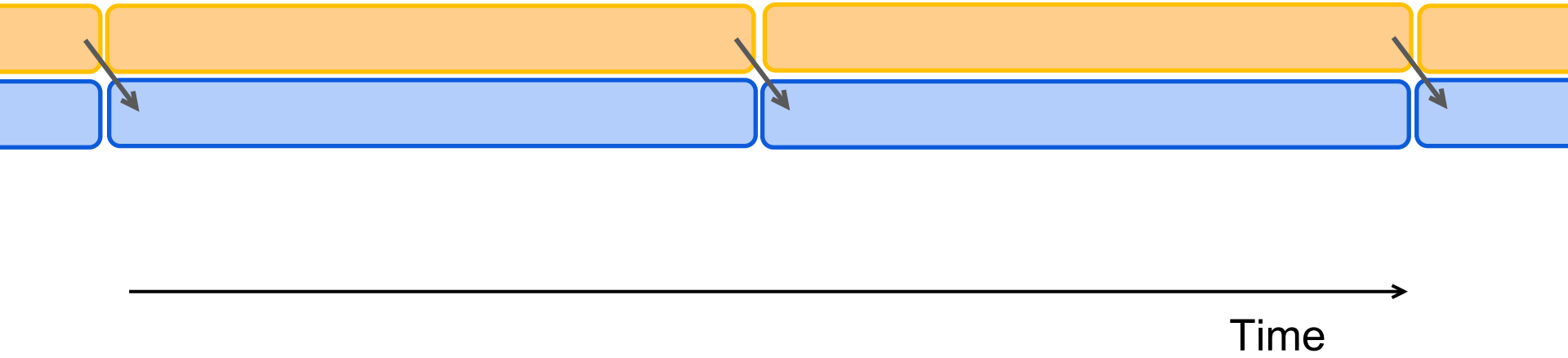


Conclusion

Asynchronous : num_worker > 0

DataLoader

Forward/Backward

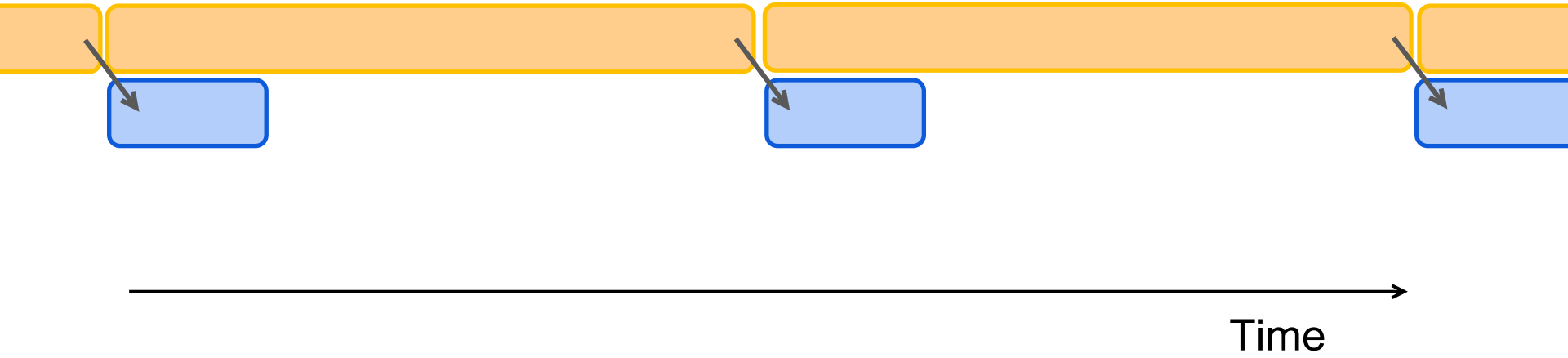


Conclusion

GPU Computing, Mixed Precision, torch.compile, ...

DataLoader

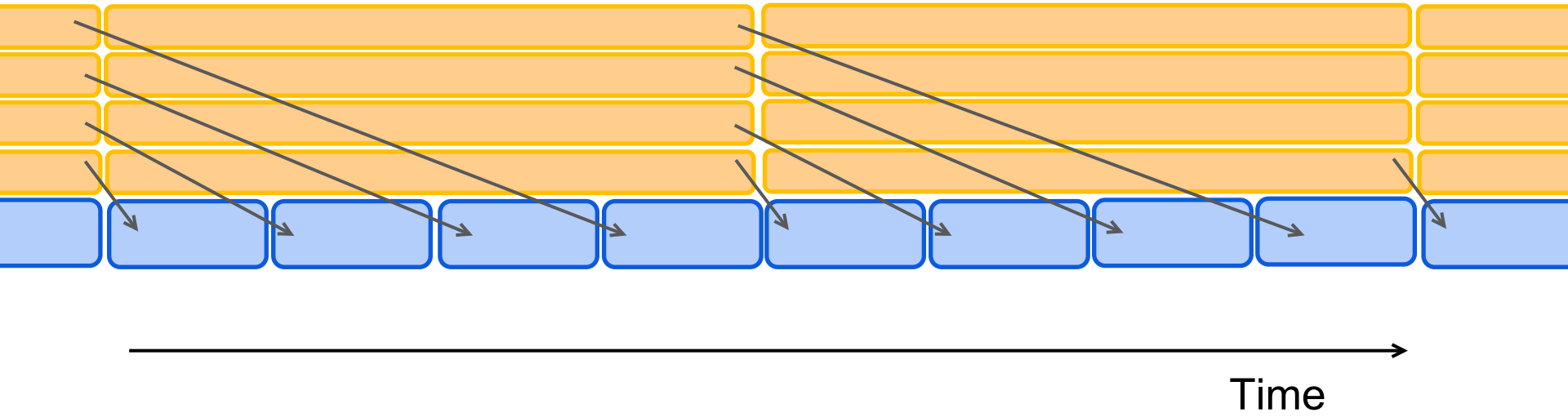
Forward/Backward



Conclusion

DataLoader Optim. : num_worker > 1, ...

DataLoader Forward/Backward



Conclusion

Training take too long !!!



Increase your batch size

Conclusion

Training take too long !!!



Increase your batch size

CUDA Out Of Memory !!!



Decrease your batch size

Conclusion



For Small Model !!!

10s or 100s M Params

Distributed Data Parallelism

Conclusion



For Small Model !!!

10s or 100s M Params

Distributed Data Parallelism

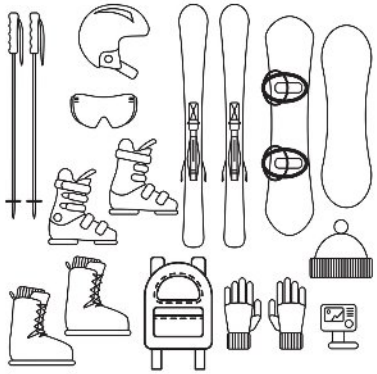


⟹ Large Batch !!

Conclusion

SGD AdamW

Weight Decay



Batch Scheduler

LR Scheduler

LR scaling

Warmup LR Decay



Conclusion



For Large Model !!!

> 1G Params

ZeRO

FSDP

Model Parallelisms

Pipeline Parallelism

Tensor Parallelism